

EFFINGHAM, IL, USA

WMO: 744661

Lat: 39.070N

Lon: 88.534W

Elev: 587

StdP: 14.39

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.9	(c) 9.9	(d) -4.4	(e) 4.4	(f) 7.1	(g) 0.4	(h) 5.7	(i) 11.4	(j) 25.6	(k) 35.2	(l) 23.3	(m) 29.4	(n) 6.2	(o) 290	(p) 0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	93.2	75.1	90.8	74.7	88.6	74.0	79.0	88.0	77.6	86.5	76.3	84.7	7.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.7	142.3	84.2	75.0	134.2	82.3	73.3	126.7	80.8	43.0	88.5	41.4	86.3	40.0	84.8	97.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 20.7	(b) 18.4	(c) 16.4	(e) -2.0	(f) 97.8	(g) 6.1	(h) 5.0	(i) -6.4	(j) 101.4	(k) -9.9	(l) 104.3	(m) -13.3	(n) 107.1	(o) -17.8	(p) 110.7		
(4) 20.7	(5) 18.4	(6) 16.4	(7) -2.0	(8) 97.8	(9) 6.1	(10) 5.0	(11) -6.4	(12) 101.4	(13) -9.9	(14) 104.3	(15) -13.3	(16) 107.1	(17) -17.8	(18) 110.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.0	29.6	32.9	44.7	55.4	65.3	73.9	76.6	75.0	68.8	56.6	44.2	35.0
	DBStd	18.80	11.76	11.41	11.23	9.30	8.45	5.70	5.34	5.22	7.08	9.66	9.81	9.90
	HDD50	2160	637	489	241	51	3	0	0	0	39	227	473	
	HDD65	4941	1096	899	634	305	106	6	1	3	41	297	624	929
	CDD50	3972	6	10	77	214	476	717	823	776	564	245	55	9
	CDD65	1277	0	0	5	18	114	273	359	314	155	38	1	0
	CDH74	12201	0	0	30	187	979	2604	3603	2894	1526	368	10	0
	CDH80	4637	0	0	1	23	275	999	1522	1128	579	110	0	0
Wind	WSAvg	6.4	8.4	8.0	8.1	8.2	6.3	5.2	4.2	3.8	4.1	6.0	7.2	7.7
Precipitation	PrecAvg	39.2	2.1	2.2	3.7	3.5	3.9	4.1	4.1	2.7	3.1	2.6	3.5	3.3
	PrecMax	52.8	5.5	5.1	7.7	7.7	8.1	12.2	9.6	6.7	8.6	7.8	10.4	9.9
	PrecMin	27.4	0.0	0.4	0.9	0.9	0.9	0.6	0.7	0.4	0.2	0.4	0.4	0.3
	PrecStd	6.7	1.5	1.2	1.9	1.8	1.9	2.5	2.3	1.6	2.3	1.5	2.2	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.8	67.4	78.7	82.3	89.8	94.1	99.2	96.8	93.4	88.5	74.7	64.3
		MCWB	59.2	59.3	63.1	64.4	72.1	74.8	74.8	76.6	71.5	72.2	61.9	58.3
	2%	DB	57.4	61.4	72.4	79.3	85.9	91.0	93.3	92.0	90.3	82.3	69.6	59.2
		MCWB	53.5	56.5	60.8	63.2	70.3	74.0	76.3	75.5	71.7	68.7	60.4	55.7
	5%	DB	53.6	55.5	67.6	75.2	83.1	89.0	90.7	89.6	87.0	77.8	64.3	54.7
		MCWB	49.6	50.4	58.0	61.6	69.0	73.6	76.2	75.6	71.0	64.7	57.5	50.5
	10%	DB	46.4	50.2	63.3	72.2	80.5	86.1	88.1	86.2	82.4	73.0	60.8	50.1
		MCWB	42.9	45.7	56.3	60.2	67.4	71.8	75.0	73.8	68.4	62.6	53.6	45.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.0	61.7	65.2	68.7	75.3	79.9	81.2	81.1	77.1	74.0	66.4	60.2
		MCDB	63.1	65.1	74.3	78.1	83.8	89.0	90.1	90.6	84.9	82.4	72.8	63.0
	2%	WB	55.2	57.0	62.5	66.0	72.9	77.3	79.2	78.4	74.8	70.8	62.4	56.1
		MCDB	57.0	60.8	70.1	74.5	81.6	86.7	88.5	87.1	83.7	79.1	66.3	58.9
	5%	WB	49.8	51.3	60.0	63.9	71.1	75.4	77.9	77.1	73.1	68.2	58.9	51.1
		MCDB	52.4	55.0	66.0	72.0	79.6	84.3	86.9	85.7	81.9	75.0	64.1	53.8
	10%	WB	42.8	45.6	57.0	61.9	69.3	73.9	76.6	75.5	71.2	64.7	53.9	46.2
		MCDB	47.0	50.2	62.6	70.0	77.3	82.7	85.2	83.3	79.8	70.9	59.0	48.8
Mean Daily Temperature Range	5% DB	MDBR	15.3	16.0	18.1	20.4	19.5	19.9	19.3	19.8	22.9	21.4	17.7	14.1
		MCDBR	21.9	23.5	23.5	24.9	22.5	22.5	21.6	22.4	26.3	25.7	23.2	19.0
		MCWBR	18.4	18.5	14.4	13.4	10.1	9.3	8.3	9.4	10.1	12.2	15.3	15.7
	5% WB	MCDBR	20.4	22.8	20.0	21.4	19.2	19.3	19.1	19.4	21.8	21.0	20.3	17.8
		MCWBR	18.9	19.4	13.6	13.6	9.7	9.3	8.7	9.5	9.9	11.3	15.6	16.4
Clear-Sky Solar Irradiance	taub		0.305	0.321	0.356	0.399	0.436	0.436	0.462	0.462	0.415	0.360	0.317	0.304
	taud		2.456	2.396	2.370	2.248	2.200	2.233	2.155	2.144	2.265	2.417	2.488	2.474
	Ebn at Noon		276	286	285	278	269	268	260	256	262	268	269	268
	Edh at Noon		25	31	35	43	46	44	47	47	39	30	25	23
All-Sky Solar Radiation	RadAvg		595	854	1168	1538	1741	2015	1961	1767	1481	1054	706	505
	RadStd		63	90	104	91	152	132	123	107	103	110	76	56

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140

Nomenclature: See separate page